

PRATIK INGLE, PhD

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Computer Science PhD in robotics and embodied AI, specialising in robot learning, perception and control for manipulation. Skilled in reinforcement and imitation learning, self-supervised representation learning (MAE, JEPA), foundation-model fine-tuning, computer vision, and multimodal and tactile perception. Experienced in building ML systems end to end: multimodal data pipelines, simulation-based synthetic data generation in MuJoCo/MJX, experiment design and evaluation, hardware prototyping, and deployment on real robots with closed-loop control. Hands-on with LLM agents (tool calling, MCP) and RAG with retrieval evaluation on the Anthropic, OpenAI and Gemini APIs and Snowflake Cortex. Strong in Python, PyTorch, OpenCV; comfortable owning open-ended problems from prototype to product and explaining results to non-specialists.

PROFESSIONAL EXPERIENCE

Doctoral Researcher

REAL Lab, IT University of Copenhagen

Copenhagen, Denmark

Oct 2023 – Present

- Built a robotic manipulation-surface platform from concept to working prototype, owning multimodal data collection, perception, control, RL policy training, sim-to-real, evaluation and the hardware underneath; produced peer-reviewed results at ICRA, RoboSoft and CASE.
- Designed the experiments and evaluation protocols behind each result: controlled benchmarks across objects of differing mass, shape and material, compared against state-of-the-art baselines, so that the claims held up under peer review.
- Set up and maintained the lab's shared Docker and GPU workstation infrastructure, standardising environments and removing repeated setup failures across researchers.
- Presented results at five international conferences, translating technical findings for audiences of up to hundreds of researchers from outside the field.

Visiting Researcher

CRISP Lab, University College London

London, United Kingdom

Mar 2026 – Present

- Built a manipulation-data pipeline that retargets egocentric human hand pos to leapXELA hand and robot actions for in-hand multi-object manipulation, including cleaning and data augmentation.
- Built a synthetic data pipeline in MuJoCo simulation twin of a three-channel (normal, shear-x, shear-y) tactile sensor that generated labelled training data at scale where physical collection was slow.
- Trained self-supervised representation models (Masked Autoencoder, JEPA) to learn embeddings from tactile signals and infer in-hand object shape and quantity, reaching over 95% accuracy and generalising across six objects.
- Optimized policy for downstream tasks such as real-time object inference with ongoing work to publish in IEEE RA-L.

Graduate Researcher

Swarm Lab (NJIT) and MOON Lab (IISER Bhopal)

Bhopal, India & Newark, USA

2020 – 2023

- Developed MOD, a depot-relocation algorithm combining weighted-centroid estimation with potential-field planning, improving multi-robot foraging throughput by 70% over a static depot (ANTS 2024).
- Showed through a computational modelling study that dynamic individual memory raises consensus rates in changing environments (Master's thesis, NJIT / IISERB).
- Delivered a remotely operated UV-C disinfection UAV to frontline COVID-19 volunteers.

Teaching Assistant

IT University of Copenhagen and IISER Bhopal

Copenhagen, Denmark & Bhopal, India

2021 – 2026

- Ran lab sessions across four technical courses (Deep Learning, Intelligent Robotics, Computer Vision, Digital Image Processing), debugging students' models, code and hardware.

Member

Indian Space Research Organization (ISRO)

India & Bhopal, India

2020 - 2021

- Developed a preliminary mission plan for conducting scientific experiments on Callisto, a Jovian moon.

AI ENGINEERING PROJECTS

- **Time Weaver** — time-tracking web application with an LLM agent integration: exposed time entries, categories and daily tasks as agent-callable tools so a model can query, filter and summarise live data on request over MCP; used to analyse months of my own productivity data.
- **RAG over research papers** — LangChain pipeline: 10 papers chunked and embedded into FAISS / Chroma; dense, BM25 and hybrid retrieval benchmarked on a hand-labelled question set, with hybrid lifting chunk-level Hit@5 from 0.66 to 0.78 (paper-level MRR 0.97).
- **Sales-intelligence agent on Snowflake Cortex** — semantic view over structured sales metrics (Cortex Analyst) and a Cortex Search service over conversation transcripts, wired as tools to a Cortex Agent with orchestration instructions; evaluated and iterated using Snowflake's agent observability, and exposed to Cursor through a Snowflake MCP server.
- **Transformer from scratch** — implemented the NLP stack in PyTorch from tokenisation through multi-head attention, as a first-principles study of the architecture behind current LLMs.

PUBLICATIONS

- **Suspended-Fabric Robotic Manipulation Surfaces: Modelling, Design Space, and Multimodular Extension**, Under review for IEEE International Conference on Robotics & Automation (ICRA 2027), South Korea.
- **Survey of robotic manipulation surfaces: Challenges and opportunities**, Under review for International Journal of Robotics Research.
- **Scalable, Distributed Control of a Manipulation Surface with Emergent Properties**, 18th International Conference on the Simulation of Adaptive Behavior (SAB 2026), Germany.
- **Scalable Surface-Based Manipulation Through Modularity and Inter-Module Object Transfer**, arXiv preprint: [arXiv:2601.21884](https://arxiv.org/abs/2601.21884), 2026.
- **Heterogeneous Object Manipulation on Nonlinear Soft Surface Through Linear Controller**, IEEE International Conference on Automation Science and Engineering (CASE), 2025. doi:10.1109/CASE58245.2025.11163787, USA.
- **Soft Manipulation Surface With Reduced Actuator Density for Heterogeneous Object Manipulation**, IEEE International Conference on Soft Robotics (RoboSoft), 2025 — selected for oral presentation. doi:10.1109/RoboSoft63089.2025.11020841, Switzerland.
- **Moving Depot(MOD): An Efficient Depot Motion Strategy for Multi-Robot Foraging**, 14th International Conference, ANTS 2024, Germany.

EDUCATION

PhD in Computer Science*IT University of Copenhagen (ITU)*

Robotics, Evolution and Art Lab (REAL)

Thesis: Learning and control for grasp-free manipulation robots

*Supervisors: Kasper Støyer & Andrés Faña**Copenhagen, Denmark**Oct 2023 – Oct 2026 (expected)***PhD External Stay***University College London (UCL)*

CRISP Lab

*Host: Lorenzo Jamone**London, United Kingdom**Mar 2026 – Aug 2026***Integrated BS–MS in Electrical Engineering and Computer Science***Indian Institute of Science Education and Research Bhopal (IISERB)*

Minor in Data Science and Engineering

Master's thesis (with Swarm Lab, NJIT): Bio-inspired collective robotics — multi-robot foraging and division of labour

Coursework: Data Science and Machine Learning, Deep Learning, Computational Linguistics, Algorithms, Control Systems

*Bhopal, India**2017 – 2023***Selected training:** Building AI Agents with Snowflake (Snowflake / Coursera, 2026); Advanced Course on Data Science & Machine Learning (2025); Deep Learning Specialization; PRAIRIE/MIAI AI Summer School.**Community/Volunteer:**

- NeurIPS Thirty-Fifth Annual Conference on Neural Information Processing Systems 2021
- International Conference for Artificial Life (ALife) 2024.

TECHNICAL SKILLS

LLMs & agents	Anthropic, OpenAI and Gemini APIs; tool / function calling, agent-callable tool design, structured outputs, prompt design and evaluation; LangChain, RAG (chunking, embeddings, FAISS / Chroma vector stores, hybrid BM25 + dense retrieval, retrieval evaluation); Snowflake Cortex Agents, Cortex Analyst and Cortex Search; Model Context Protocol; transformer internals implemented from scratch in PyTorch
Machine learning	PyTorch, TensorFlow, self-supervised representation learning (Masked Autoencoders, JEPA), embeddings, transformers, CNNs, reinforcement and imitation learning, multimodal data
Data & evaluation	Experiment design, statistical analysis, evaluation metrics and benchmarking against baselines, synthetic data generation pipelines, R, MATLAB
Software & MLOps	Python, C; Git, Docker, Linux, GPU workstation and remote-development setup, web application development
Robotics	MuJoCo (MJX, parallel simulation), ROS, closed-loop control, tactile sensing, embedded hardware and rapid prototyping

LINGUISTIC SKILLS

English	Full professional proficiency
Hindi	Native